

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CL, ME)

April/May 2012

CL202 Strength of Materials

Date: 02.05.2012, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) State the assumptions made in the theory of simple bending. [03]
(b) Determine the moment of inertia of the shaded area as shown in **Figure 1** about XX axis. [04]
- Q - 2 (a) A rectangular beam 60 mm wide and 150 mm deep is simply supported over a span of 6 m. If the beam is subjected to central point load of 12 kN, find the maximum bending stress induced in the beam section. [03]
(b) State and prove the perpendicular axis theorem. [03]
(c) An I-section is made up of three rectangles as shown in **Figure 2**. Find the moment of inertia of the section about the horizontal axis passing through the centre of gravity of the section. [04]
(d) A cantilever beam is rectangular in section having 80 mm width and 120 mm depth. If the cantilever is subjected to a point load of 6 kN at the free end and the bending stress is not to exceed 40 Mpa, find the span of the cantilever beam. [04]

OR

- Q - 2 (a) A steel wire of 5 mm diameter is bent into a circular shape of 5 m radius. Determine the maximum stress induced in the wire. Take $E = 200 \text{ GPa}$. [03]
(b) Find moment of inertia of a rectangular section with width b and depth d . [03]
(c) Find moment of inertia of the shaded area @ the base of a section. (Refer **Figure 3**) [04]
(d) Derive $\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$, with usual notations. [04]
- Q - 3 Attempt any Two [14]
(a) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 4**.
(b) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 5**.
(c) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 6**.

SECTION - II

- Q - 4 (a) A wooden beam 100 mm wide, 250 mm deep and 3 m long is carrying a uniformly distributed load of 40 kN/m. Determine the maximum shear stress and sketch the [03]

variation of shear stress along the depth of the beam.

- (b) Define torsion and state the assumptions made for derivations of general equation of torsion. [04]

Q - 5 (a) In a close coiled helical spring, the diameter of each coil is to be 10 times that of wire of the spring and the maximum shear stress is not to exceed 60 N/mm^2 . Maximum permissible deflection under a load of 500 N is 10 cm. Determine the number of coils, the diameter of the coil and energy stored in the coil. Take shear modulus as $9 \times 10^4 \text{ N/mm}^2$. [07]

- (b) The section of a beam shown in **Figure 7** is subjected to a shear force of 200 kN. Determine the shear stress at significant points and sketch the shear stress distribution diagram. [07]

OR

Q - 5 (a) 450 kW power has to be transmitted at 100 rpm. Find (i) the necessary diameter of a solid circular shaft, (ii) the necessary diameter of a hollow circular section, the diameter being $\frac{3}{4}$ of external diameter. Allowable shear stress = 75 MPa. Density of material 77 kN/m^3 . Which is more economical? [07]

- (b) Derive $q = \frac{SAY}{BI}$ with usual notations. [07]

Q - 6 Attempt any Two [14]

- (a) A point in a strained material is subjected to normal and shear stress as shown in **Figure 8**. Determine Normal (p_n), Tangential (p_t) and Resultant (p_r) stresses on a plane inclined at 60° with vertical as shown in **Figure 8**. Also find major and minor principal stresses and maximum tangential stress ($p_{t \max}$) and its location.

- (b) A point in a strained material is subjected to normal and shear stress as shown in **Figure 9**. Determine Normal (p_n), Tangential (p_t) and Resultant (p_r) stresses on a plane inclined at 60° with vertical as shown in **Figure 9**. Also find major and minor principal stresses and maximum tangential stress ($p_{t \max}$) and its location.

- (c) Define principal plane and state the importance of finding the same. Draw Mohr's stress circle for the element subjected to the stresses as shown in **Figure 10** and find: (i) Major and minor principal stresses (ii) location of principal planes and (iii) maximum shear stress and its location.

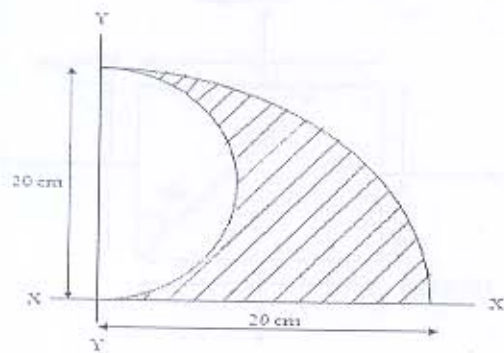


Figure 1

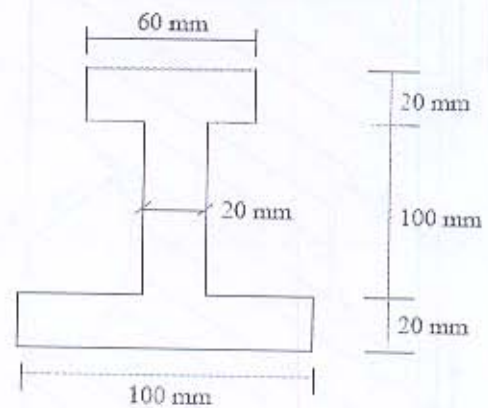


Figure 2

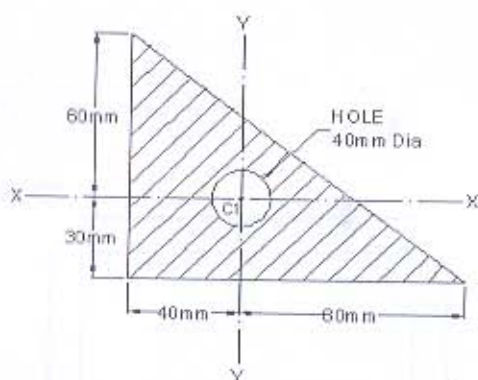


Figure 3

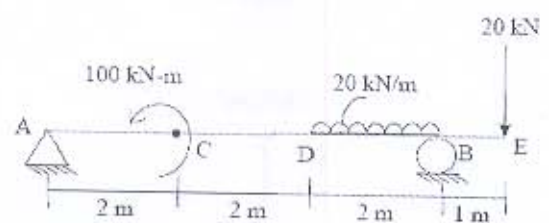


Figure 4

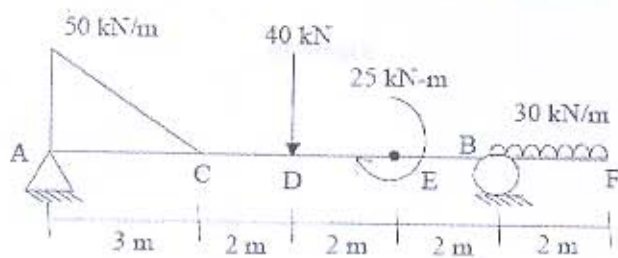


Figure 5

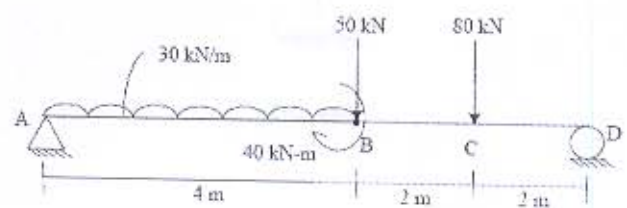


Figure 6

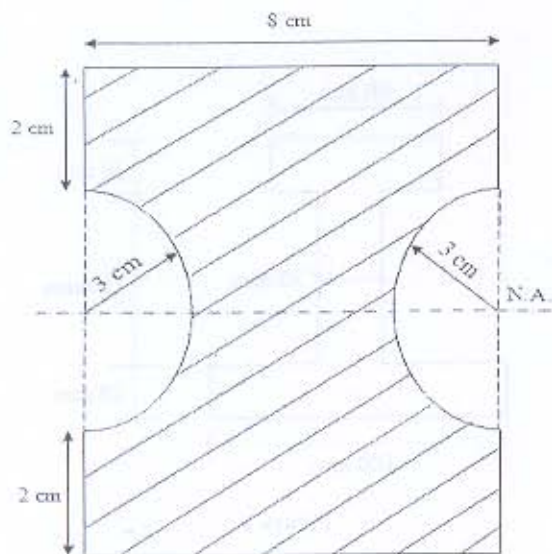


Figure 7

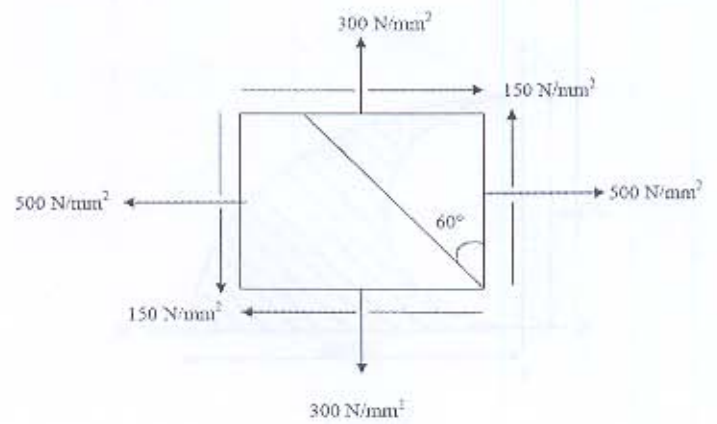


Figure 8

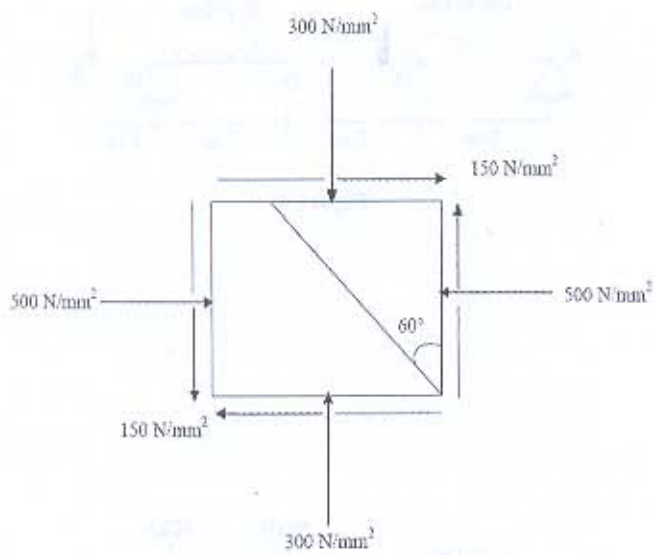


Figure 9

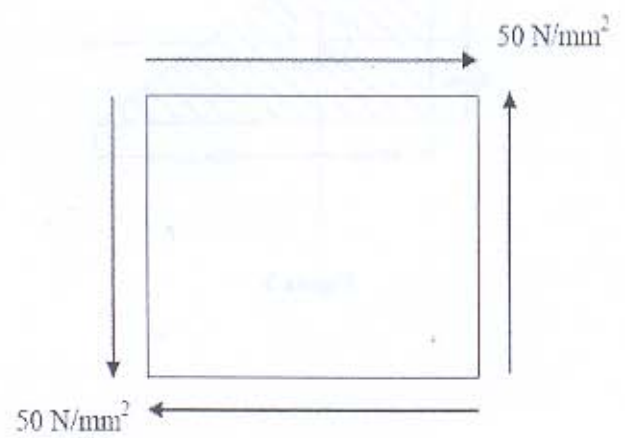


Figure 10
